

Assignment 1

In this assignment, you will implement various image classification models, based on the SVM / Softmax / Two-layer Neural Network. The goals of this assignment are as follows:

- Implement and apply a Multiclass Support Vector Machine (SVM) classifier
- Implement and apply a Softmax classifier
- Implement and apply a Two-layer Neural Network classifier
- Understand the differences and tradeoffs between these classifiers
- Understand how a Two-layer Neural Network can approximate an arbitrary function
- Practice implementing vectorized gradient code by checking against naive implementations, and using numeric gradient checking

Q1: Linear Classifiers

The notebook **linear_classifier.ipynb** will walk you through implementing SVM and Softmax classifier. You are required to write code on **linear_classifier.py**.

Q2: Two-layer Neural Network

The notebook **two_layer_net.ipynb** will walk you through implementing a two-layer neural network-based classifier. Your implementation will go to **two_layer_net.py**.

Steps

1. Download the zipped assignment file

Could be found in moodle

2. Unzip all and open the Colab file from the Drive

Once you unzip the downloaded content, please upload the folder to your Google Drive. Then, open each `*.ipynb` notebook file with Google Colab by right-clicking the `*.ipynb` file. We recommend editing your `*.py` file on Google Colab, set the ipython notebook and the code side by side. For more information on using Colab, please see [Colab tutorial](#).

3. Work on the assignment

Work through the notebook, executing cells and writing code in `*.py`, as indicated. You can save your work, both `*.ipynb` and `*.py`, in Google Drive (click "File" -> "Save") and resume later if you don't want to complete it all at once.

While working on the assignment, keep the following in mind:

- The notebook and the python file have clearly marked blocks where you are expected to write code. **Do not write or modify any code outside of these blocks.**
- **Do not add or delete cells from the notebook.** You may add new cells to perform scratch computations, but you should delete them before submitting your work.
- **Run all cells, and do not clear out the outputs, before submitting.** You will only get credit for code that has been run.

4. Submission

The following instructions are mandatory and will be checked and graded by the course staff. Failing to follow these instructions will reduce points from your grade. The assignment is to be done in Pytorch and submitted to the course moodle page in the form of a *.zip (not RAR) containing your implementation in python of all the parts (2 notebooks with your results at each required cell, and 2 .py files).

The ZIP file name should be the initials and ID of both of the team members ex. 'TB-1234567_RS-7654321.zip'.

For any question regarding this assignment please refer to the course forum on the moodle web site, for personal questions only please email shachart@hit.ac.il